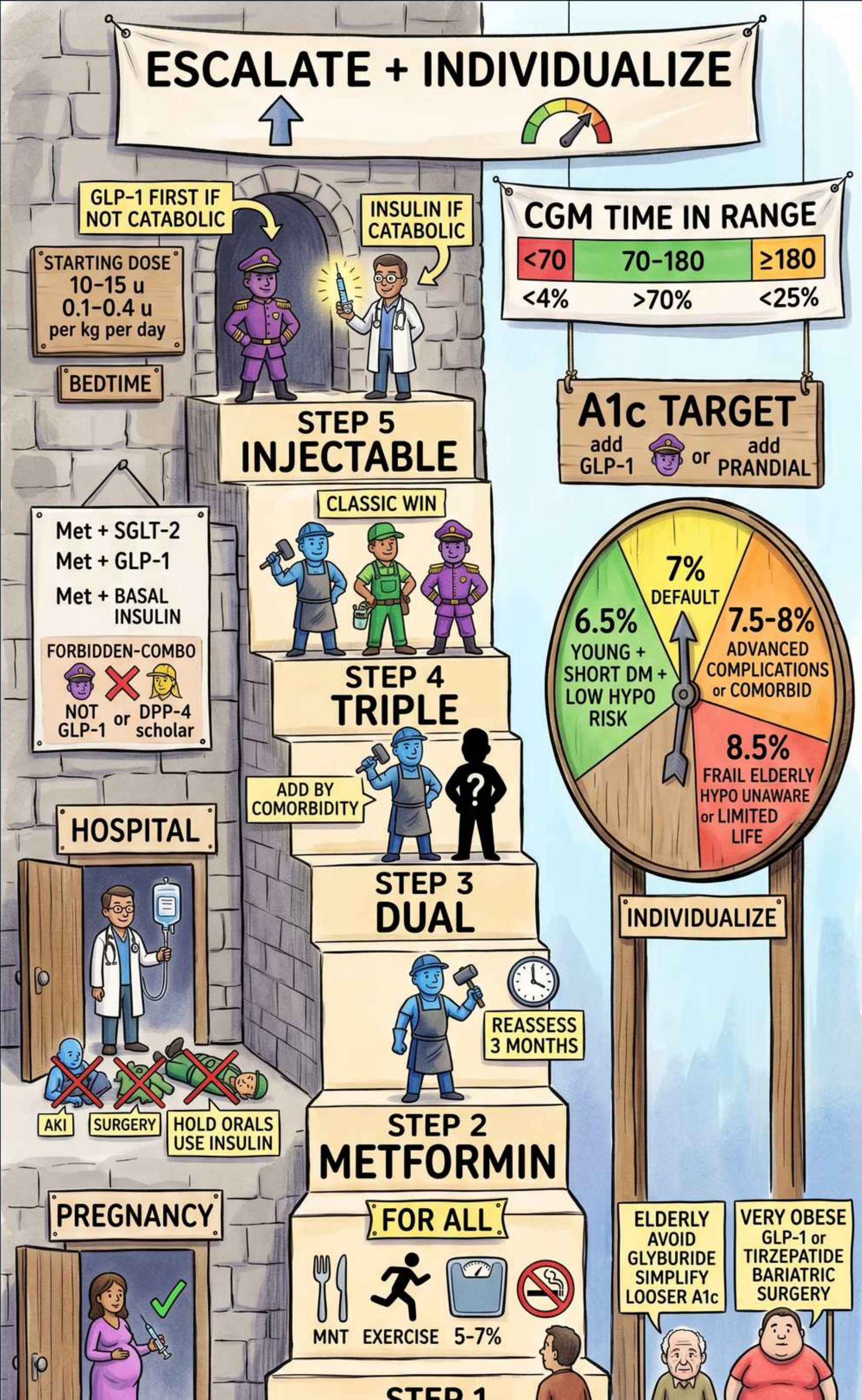




1. Step 1 Lifestyle

WHAT YOU SEE

Bottom step STEP 1 LIFESTYLE, MNT, exercise, target 5-7%

WHAT IT ENCODES

Step 1 for ALL patients: medical nutrition therapy, ≥ 150 min/week moderate aerobic exercise, and 5-7% body-weight loss. Lifestyle continues at every subsequent step, not just at diagnosis. Even modest weight loss improves insulin sensitivity and can lower A1c by ~ 1 -2%.

BOARD TRAP

WRONG: 'Start metformin alone and skip lifestyle counseling once a drug is on board.' Lifestyle modification is foundational and continues lifelong alongside every pharmacologic step. Discriminator: drugs are added TO lifestyle, never instead of it.

2. Step 2 Metformin

WHAT YOU SEE

STEP 2 METFORMIN FOR ALL, reassess 3 months

WHAT IT ENCODES

Step 2: metformin (start 500 mg daily with meals, titrate to 1000 mg BID) is first-line for essentially all T2DM unless contraindicated. Mechanism: decreases hepatic gluconeogenesis (AMPK), weight-neutral, no hypoglycemia. Reassess A1c at ~ 3 months. Hold if eGFR < 30 (lactic-acidosis risk).

BOARD TRAP

WRONG: 'A patient with new T2DM and A1c 7.4% should start a sulfonylurea because it lowers glucose fastest.' Metformin is first-line for nearly everyone; sulfonylureas cause hypoglycemia and weight gain and are not preferred initial therapy. Discriminator: default monotherapy = metformin unless contraindicated.

3. Step 3 Dual therapy

WHAT YOU SEE

STEP 3 DUAL, add by comorbidity

WHAT IT ENCODES

Step 3: add a second agent chosen by COMORBIDITY, not by A1c alone. ASCVD/high CV risk $\rightarrow$ GLP-1 RA (or SGLT2i with proven CV benefit). HFrEF/HFpEF or CKD $\rightarrow$ SGLT2i. These benefits apply REGARDLESS of baseline A1c, even if A1c is already at goal on metformin.

BOARD TRAP

WRONG: 'A T2DM patient with HFrEF at A1c 6.8% on metformin needs no additional drug because they are at goal.' An SGLT2i is indicated for the heart-failure benefit independent of glycemic control. Discriminator: pick the second agent by ASCVD/HF/CKD comorbidity, not by whether A1c is already $< 7\%$.

4. Step 4 Triple therapy

WHAT YOU SEE

STEP 4 TRIPLE

WHAT IT ENCODES

Step 4: triple non-insulin combination (e.g., metformin + SGLT2i + GLP-1 RA) if still above target after dual therapy. Choose agents with complementary mechanisms and avoid overlapping/contraindicated pairs. Prioritize agents with cardiorenal benefit and low hypoglycemia risk.

BOARD TRAP

WRONG: 'Add a sulfonylurea as the third agent to a patient already on a GLP-1 RA without adjusting it.' Stacking insulin-secretagogues raises hypoglycemia risk; doses must be reduced and weight-neutral/cardioprotective agents are preferred. Discriminator: build triple therapy around complementary, low-hypoglycemia agents.

5. Step 5 Injectable

WHAT YOU SEE

STEP 5 INJECTABLE, GLP-1 first if not catabolic

WHAT IT ENCODES

Step 5: when injectable therapy is needed, prefer a GLP-1 RA BEFORE basal insulin (greater A1c reduction, weight loss, CV benefit, less hypoglycemia) UNLESS the patient is catabolic. GLP-1 RAs and dual GIP/GLP-1 agonists (tirzepatide) are potent A1c-lowering injectables.

BOARD TRAP

WRONG: 'Any T2DM patient needing an injectable should start basal insulin first.' Non-catabolic patients should generally get a GLP-1 RA before insulin. Discriminator: insulin-first is reserved for catabolic presentations (see next), otherwise GLP-1 RA is preferred.

6. Insulin if catabolic

WHAT YOU SEE

INSULIN IF CATABOLIC, starting dose 0.1-0.4 u/kg bedtime

WHAT IT ENCODES

Start INSULIN promptly if catabolic features are present: unintentional weight loss, ketosis/ketonuria, symptoms of marked hyperglycemia, A1c $> 10\%$ or glucose ≥ 300 , or to distinguish from possible T1DM/LADA. Basal insulin start: 0.1-0.2 u/kg/day (up to ~ 0.4) at bedtime, titrate by fasting glucose.

BOARD TRAP

WRONG: 'A newly diagnosed lean patient with A1c 11%, ketonuria, and weight loss should start metformin and recheck in 3 months.' Catabolic/symptomatic hyperglycemia requires INSULIN now (and consider T1DM/LADA workup). Discriminator: ketosis + weight loss + very high A1c = insulin first, not a slow oral titration.

7. Forbidden combo: no GLP-1 + DPP-4

WHAT YOU SEE

FORBIDDEN-COMBO box: NOT GLP-1 + DPP-4

WHAT IT ENCODES

Do NOT combine a GLP-1 RA with a DPP-4 inhibitor: both act on the incretin (GLP-1) axis, so DPP-4i adds no glycemic benefit on top of a GLP-1 RA and only adds cost/side effects. If starting a GLP-1 RA, stop the DPP-4i.

BOARD TRAP

WRONG: 'Add sitagliptin (DPP-4i) to a patient already on semaglutide (GLP-1 RA) for extra incretin effect.' These share the incretin pathway and should not be combined. Discriminator: GLP-1 RA + DPP-4i = redundant; pick one incretin-based agent.

8. Preferred combos w/ SGLT2 & GLP-1

WHAT YOU SEE

Met+SGLT2, Met+GLP-1, Met+basal insulin combos

WHAT IT ENCODES

Preferred comorbidity-driven pairings: metformin + SGLT2i (HF/CKD benefit), metformin + GLP-1 RA (ASCVD benefit, weight loss), and metformin + basal insulin when insulin is needed. SGLT2i + GLP-1 RA can be combined (different mechanisms) and gives additive cardiorenal protection.

BOARD TRAP

WRONG: 'SGLT2i and GLP-1 RA cannot be used together because both are newer agents.' They act by distinct mechanisms (glycosuria vs incretin) and are commonly combined for additive cardiorenal benefit. Discriminator: the only forbidden incretin overlap is GLP-1 RA + DPP-4i, not SGLT2i + GLP-1 RA.

9. A1c target 7% default dial

WHAT YOU SEE

Color dial 7% DEFAULT

WHAT IT ENCODES

Default glycemic target is A1c <7% for most non-pregnant adults (ADA). This corresponds to an estimated average glucose ~154 mg/dL. The target is individualized up or down based on hypoglycemia risk, life expectancy, duration, and comorbidities.

BOARD TRAP

WRONG: 'Aim for A1c <7% in every diabetic regardless of age or frailty.' The <7% default must be loosened for frail/elderly or tightened for low-risk young patients. Discriminator: <7% is the DEFAULT, not a universal target.

10. Tighter 6.5% (young/short DM)

WHAT YOU SEE

6.5% young + short DM + low hypo risk

WHAT IT ENCODES

A tighter target (A1c <6.5%) is appropriate for younger patients, short DM duration, long life expectancy, no significant CVD, and low hypoglycemia risk — when it can be achieved without undue hypoglycemia or polypharmacy burden.

BOARD TRAP

WRONG: 'Push a frail 82-year-old on a sulfonylurea to A1c <6.5% for maximal protection.' Tight targets are for low-risk YOUNG patients; in the elderly this causes dangerous hypoglycemia with little benefit. Discriminator: <6.5% requires low hypoglycemia risk and long life expectancy.

11. Looser 7.5-8.5% (frail/elderly)

WHAT YOU SEE

7.5-8% advanced complications/comorbid; 8.5% frail elderly/hypo unaware/limited life

WHAT IT ENCODES

Looser targets: A1c 7.5-8% for advanced microvascular complications, extensive comorbidity, or longstanding hard-to-control DM; up to ~8-8.5% for frail elderly, hypoglycemia unawareness, or limited life expectancy. Avoiding hypoglycemia outweighs marginal glycemic gains here.

BOARD TRAP

WRONG: 'A nursing-home patient with dementia and hypoglycemia unawareness needs intensification to reach A1c 7%.' Such patients warrant a relaxed target (~8-8.5%) to prevent hypoglycemia. Discriminator: frailty/limited life expectancy/hypo unawareness -> loosen, do not intensify.

12. CGM time-in-range

WHAT YOU SEE

CGM TIME IN RANGE 70-180 >70%, <70 <4%, >=180 <25%

WHAT IT ENCODES

Continuous glucose monitoring (CGM) metrics: time-in-range (70-180 mg/dL) >70%; time-below-range (<70) <4% and (<54) <1%; time-above-range (>180) <25%. Each ~10% increase in time-in-range correlates with roughly a 0.5-0.8% lower A1c. A Glucose Management Indicator (GMI) estimates A1c from CGM data.

BOARD TRAP

WRONG: 'A1c is always the most reliable marker, so CGM time-in-range is unnecessary.' A1c is misleading with anemia, hemoglobinopathies, CKD, recent transfusion, or pregnancy, where CGM/time-in-range is superior. Discriminator: use CGM metrics when A1c is unreliable or for granular hypoglycemia data.

13. Pregnancy uses insulin

WHAT YOU SEE

PREGNANCY checkmark, usually insulin

WHAT IT ENCODES

In pregnancy, INSULIN is the preferred glucose-lowering therapy (does not cross the placenta, best safety data). Most non-insulin agents are avoided; GLP-1 RAs, SGLT2i, and statins are contraindicated. Glycemic goals are tighter (fasting <95, 1-h postprandial <140, 2-h <120 mg/dL).

BOARD TRAP

WRONG: 'Continue the patient's metformin/empagliflozin regimen unchanged when she becomes pregnant.' SGLT2i (and GLP-1 RA, statins) must be stopped; insulin is the standard. Discriminator: pregnancy = insulin-based therapy with tighter targets, not oral/incretin agents as first choice.

14. Special-population stand

WHAT YOU SEE

Elderly avoid glyburide, very obese GLP-1/tirzepatide/bariatric

WHAT IT ENCODES

Special populations: AVOID glyburide (long-acting sulfonylurea) in the elderly and in CKD (prolonged hypoglycemia) — it is a Beers-criteria drug. In severe obesity, favor GLP-1 RA or dual GIP/GLP-1 agonist (tirzepatide) for weight loss, and consider bariatric/metabolic surgery (esp. BMI >=35 with uncontrolled DM).

BOARD TRAP

WRONG: 'Glyburide is a safe, cheap first-line agent for an 80-year-old with mild CKD.' Glyburide accumulates in renal impairment and causes severe, prolonged hypoglycemia in the elderly. Discriminator: in elderly/CKD prefer agents with low hypoglycemia risk (metformin, SGLT2i, GLP-1 RA, DPP-4i); avoid glyburide.